



How Crime Relates to Investment and Disinvestment in Residential Properties:

A Baltimore City Case Study

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Introduction

High-crime areas are typically stuck in a spiral: disorder and decay beget crime, crime begets disorder and decay (Skogan, 1992). This disorder can show up in dilapidated infrastructure, placing residents in increasingly unsafe living conditions. Baltimore, a city with many abandoned buildings and a high crime rate, may be an example of this spiral.

Existing literature primarily examines crime as the outcome of infrastructural conditions (e.g., Broken Windows). However, there is limited research on how crime itself might explain variations in infrastructure, particularly in terms of conditions shaped by the investment choices made by individuals and municipalities.

Question: Is investment in residential properties in Baltimore responsive to crime rates?

Methods & Data

Method: OLS regression with two-way fixed effects and a vector of demographic controls.

Panel data from 2015-2021:

$y = \text{crime}$	$x = \text{residential investment}$
Part 1 Crime (total major crimes), broken down into violent and property-related, as measured by: - Incidents per 1,000 residents - Violent: homicide, rape, aggravated assault, and robbery - Property: burglary, larceny, and auto theft	Model 1: residential investment as <i>rehabilitation</i> - The % of residential properties with a permit to renovate where the cost exceeds \$5,000 Model 2: residential investment as <i>demolition</i> - The # of demolition permits per 1,000 existing residential properties

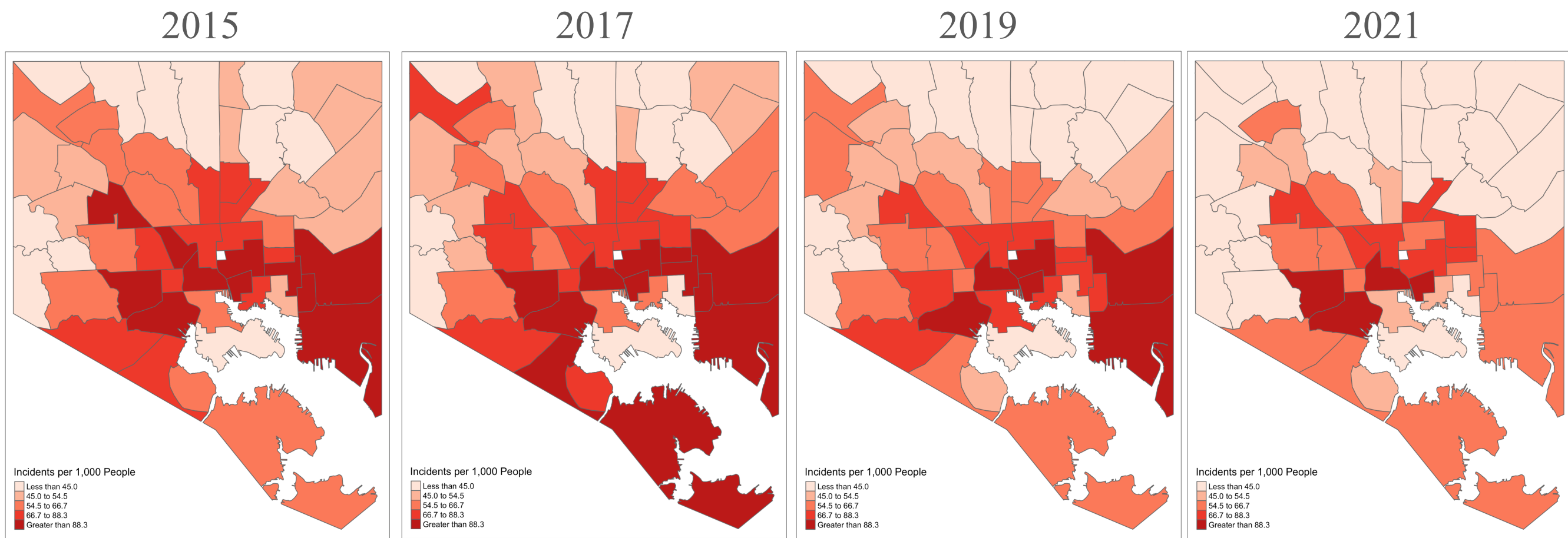
- Controls:
- Fixed effects: spatial (Community Statistical Areas (CSAs)) and temporal (years)
 - Demographic: race, ethnicity, age, educational attainment, employment status, poverty status

Figure 1. CSAs and Census Tracts



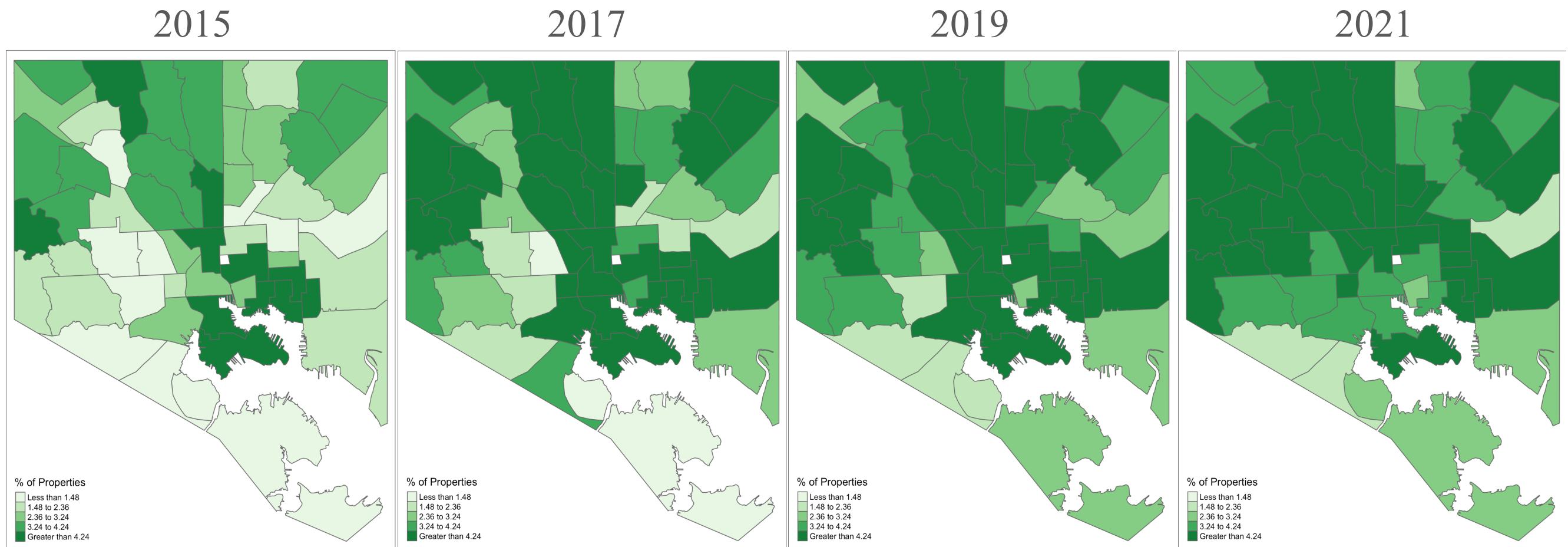
Source: Baltimore Neighborhood Indicators Alliance (2020)

Figure 2. Spatial Distribution of Part 1 Crime Rates



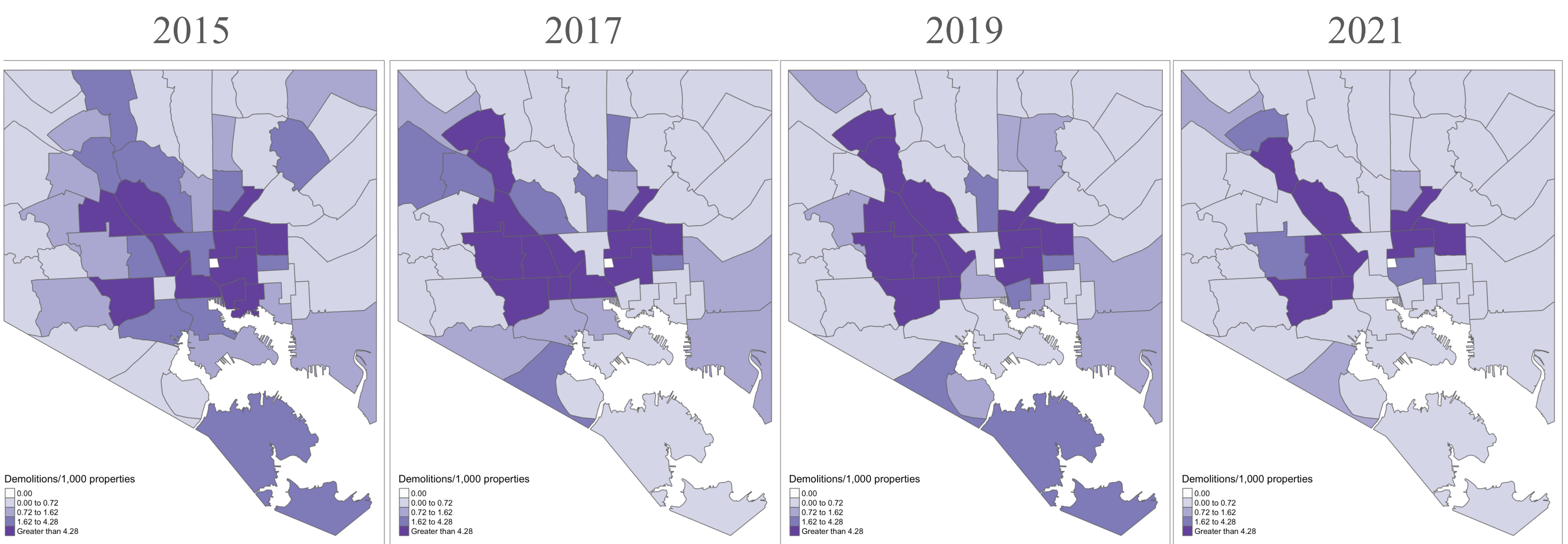
Incidents per 1,000 people. Source: Author's calculations using Baltimore Police Department data, 2015-2021

Figure 3. Increasing Residential Rehabilitations



Permits as % of residential properties. Source: Author's calculations using Baltimore Department of Housing and Community Development data, 2015-2021

Figure 4. Decreasing Residential Demolitions



Demolitions per 1,000 residential properties. Source: Author's calculations using Baltimore Department of Housing and Community Development data, 2015-2021

Equations

$$\% \text{ rehabs}_{CSA,t} = \beta_1 \text{ part 1 crime}_{CSA,t-1} + \sum_{i=2...k} \beta_i X_{CSA,t} + \gamma_{CSA} + \delta_t + \epsilon_{CSA,t} \quad (\text{Model 1})$$

$$\text{demolitions}_{CSA,t} = \beta_1 \text{ part 1 crime}_{CSA,t-1} + \sum_{i=2...k} \beta_i X_{CSA,t} + \gamma_{CSA} + \delta_t + \epsilon_{CSA,t} \quad (\text{Model 2})$$

Figure 5. Model 1: More residential rehabilitations occur in higher crime areas

Crime and Rehabilitation Permits			
	Dependent variable:		
	Rate of Rehabilitation Permits		
	(1)	(2)	(3)
Lag of Total Crime	0.008** (0.003)		
Lag of Violent Crime		0.021** (0.010)	
Lag of Property Crime			0.010* (0.005)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.087	0.084	0.081
Adjusted R ²	-0.131	-0.136	-0.140
Note:	*p<0.1; **p<0.05; ***p<0.01		

Notes: Standard errors are in parentheses. Levels of statistical significance are indicated at * p<0.1, ** p<0.05, *** p<0.01. 384 observations are used from the full panel of 385 observations due to the one-year lag on crime. Results rendered with stargazer (Hlavac, 2022). Source: Author's calculations using Baltimore Department of Housing and Community Development, Baltimore Police Department, and US Census American Community Survey data, 2015-2021.

Results

Investment in residential properties has a significant relationship to the crime rates in Baltimore. Specifically:

- Model 1:
- ↑ crime, ↑ rehabilitations
- Model 2:
- ↑ crime, ↓ demolitions

This suggests that residents themselves are investing in their properties more frequently in higher-crime areas, while the government is lacking in its investment in these higher-crime areas.

Robustness checks:

- Varying the time lag of crime
- Additional demographic controls
- COVID-19 dummy for 2020 and 2021

The results of these alternate specifications confirm that the original specifications are appropriate.

Figure 6. Model 2: Fewer residential demolitions occur in higher crime areas

Crime and Demolitions			
	Dependent variable:		
	Rate of Demolition Permits		
	(1)	(2)	(3)
Lag of Total Crime	-0.033*** (0.012)		
Lag of Violent Crime		-0.096*** (0.032)	
Lag of Property Crime			-0.040** (0.017)
Demographic Controls	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
CSA Fixed Effects	Yes	Yes	Yes
# of CSAs	55	55	55
Observations	384	384	384
R ²	0.046	0.048	0.038
Adjusted R ²	-0.183	-0.180	-0.192
Note:	*p<0.1; **p<0.05; ***p<0.01		

Conclusion

My findings affirm the arguments advanced by existing literature which assert that crime is strongly related to indicators of physical disorder while offering a novel perspective on how crime relates to actions that may *change* conditions of physical and social disorder.

Policymakers should:

- focus more demolition efforts in high-crime neighborhoods

Researchers should:

- keep an eye on trends in residential investment given Mayor Brandon Scott's recent Violence Prevention Plan

References

Full paper and references available upon request

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